

Cartography of littoral rocky-shore communities (CARLIT) as a tool for ecological quality assessment of coastal waters in the Eastern Adriatic Sea



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ABSTRACT

Littoral cartography is a valuable tool which uses shallow water macroalgal communities for monitoring ecosystem health and water quality. Cartography of littoral rocky-shore communities (CARLIT) is a European Union Water Framework Directive-compliant monitoring method widely used in the Western Mediterranean Sea. This non-destructive method is based on a visual observation of the type and length of coast occupied by rocky-shore communities in the upper-sublittoral zone. Here we present the first wide-scale application of CARLIT in the Adriatic Sea. The reference values calculated specifically for the Croatian coast (Eastern Adriatic Sea) ensure accurate calculation of the ecological status. The measured ecological quality ratio values (EQR) show a good linear relation with the Land Uses Simplified Index (LUSI) and appropriately reflect an increase in nutrient concentrations. The results suggest that this method can be used as a reliable biomonitoring tool in the Eastern Adriatic Sea. This study highlighted that *Cystoseira* forests thrive in 39% of the surveyed coastline and 23% of the coastline seems to be subjected to overgrazing by sea urchins. Our results provide a spatially accurate information on the abundance of different upper-sublittoral communities and the associated ecological status, which can be used in future management plans for improving water quality.

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1. Introduction

The European Union Water Framework Directive (WFD) (EC, 2000) sets a legal framework for monitoring the ecological quality of European waters. It recognizes the importance of biological indicators and uses “biological quality elements” (BQE) to classify water bodies into “ecological status classes” (ESC). According to WFD, the ecological status is expressed as an “ecological quality ratio” or EQR (ranging from 0 – low quality to 1 – high quality), which is calculated as a quotient between the measured value of an indicator in a site and a reference value which is defined by incorporating the natural variation of the considered BQE (EC, 2003).

Scientific community has responded to the need for this new type of bio-monitoring tools with the development of new indices for the estimation of ecological quality of marine coastal waters (for

a review, see Marques et al., 2009; Martínez-Crego et al., 2010). Benthic macroalgae are considered valuable biological indicators of water quality worldwide (Arévalo et al., 2007; Ballesteros et al., 2007; Pinedo et al., 2007; Wells et al., 2007; Juanes et al., 2008; Marques et al., 2009). The CARLIT method (cartography of littoral rocky-shore communities) was developed in the North-Western Mediterranean (Ballesteros et al., 2007) and proposed for use, under the scope of WFD, for the BQE macroalgae. It is applied in monitoring programs in Spain (Ballesteros et al., 2007), Italy (Mangialajo et al., 2007; Asnaghi et al., 2009), Malta and France (Blanfuné et al., 2011). CARLIT is a non-destructive method, based on visual observation of the type and length of coast occupied by categorized dominant macroalgal communities in the upper-sublittoral zone. Diaz et al. (2004) pointed out that a broad-scale mapping of macroalgae distribution patterns is essential for the assessment of their status, and this kind of information is also necessary for the implementation of other important EU directives, such as Habitats and Marine Strategy Framework Directives.

A successful environmental monitoring program requires a sampling design with spatial and temporal replicates in order to reflect variability (Underwood, 1993). The time and cost of sample

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collection and laborious species identification in all the replicates usually restricts the spatial extent of the monitoring (Bates et al., 2007). Therefore, most biomonitoring methods assess the ecological status of an environment on the basis of data collected from one or few stations, under the assumption that the assessment is applicable to the whole environment (e.g. Orfanidis et al., 2001; Wells et al., 2007). Although this may generally be applicable, it comes with a great risk of underestimation of the level of heterogeneity in a given water body, which could subsequently add to uncertainty in ecological quality assessment (Clarke and Hering, 2006). A recent research shows high uncertainty in the assessment of the ecological status associated with spatial scales of macroalgal and seagrass sampling (Mascaró et al., 2012a). The CARLIT method reduces this concern by taking in consideration the exhaustive cartography of the whole coastline wherever possible or alternatively, by the hierarchical sampling whilst ensuring the results are representative of the water mass which is being considered (Ballesteros et al., 2007).

In order to evaluate its advantages and shortages, Ballesteros et al. (2007) have recommended application of the CARLIT method in other Mediterranean regions. The method has never been applied in the Eastern Mediterranean Sea, except for a very limited implementation of the modified CARLIT on anthropic structures in the Venice lagoon in the Northern Adriatic (Sfriso and Facca, 2011). Therefore, this study represents the first wide-scale application of CARLIT in the Adriatic Sea, including the revision of the communities which are to be considered and the definition of reference conditions specific to the eastern part of the Adriatic Sea. The validation of the index is done by comparing the results with the land use index commonly used in the framework of WFD and with the water quality in selected sites.

2. Methods

2.1. Adaptation of the method to the Eastern Adriatic Sea

The CARLIT methodology in this study was applied according to general procedures elaborated by Ballesteros et al. (2007). However, slight modifications were introduced to account for the specific attributes of macroalgal communities in the Adriatic Sea and a list of typical upper-sublittoral communities in the eastern part of the Adriatic Sea was drafted. This was followed by assigning a “sensitivity level” (SL), ranging from 1 to 20, to each community type (Table 1). The classification of communities and designation of their sensitivity level for the Croatian coast was based on the existing table for the North-Western Mediterranean Sea (Ballesteros et al., 2007; Asnaghi et al., 2009). In order to make it more applicable to the Adriatic Sea, an extensive scientific literature search from the Mediterranean and the Adriatic Sea was

performed (Ercegović, 1952, 1960, 1964, 1966, 1980; Golubić, 1960, 1970; Munda, 1973, 1974, 1979, 1980a,b, 1982; Belsher, 1977; Boudouresque, 1984; Chrysosvergis and Panayotidis, 1995; Špan and Antolić, 1997; Panayotidis et al., 1999; Soltan et al., 2001; Arévalo et al., 2007; Ballesteros et al., 2007; Pinedo et al., 2007; Mangialajo et al., 2008; Orlando-Bonaca et al., 2008; Iveša et al., 2009; Sliškočić et al., 2011). A workshop with experts from Croatia, Italy, Slovenia, Montenegro, Spain and France was organized in order to validate the proposed list of communities with the associated sensitivity levels and the applicability of the CARLIT in the Adriatic Sea. As proposed in the North-Western Mediterranean Sea CARLIT table, the highest SL values were assigned to well developed *Cystoseira* forests, intermediate values to degraded communities with photophilic turf algae and mussel beds, whilst the lowest values were assigned to heavily degraded communities found in extremely polluted areas, consisting mostly of green algae and/or cyanobacteria. The proposed list of communities included the most common *Cystoseira* species which can be found in the upper-sublittoral in the Adriatic Sea. The most characteristic species is *Cystoseira amentacea* var. *spicata*, which forms a visible belt in the upper-sublittoral zone of exposed zones. In order to simplify data collection and analysis, we propose that the abundance of its stands should be classified into three levels (eliminating two intermediate classes in Ballesteros et al., 2007): continuous belt, abundant patches and rare scattered plants. A lower SL value was assigned to *Cystoseira barbata*, because it is commonly observed in slightly polluted zones across the studied area. The shallow seagrass formations (for example “*Posidonia* reefs”) were not considered in the proposed list since the application of CARLIT was limited to the shallower zone of sheltered and exposed rocky shores (in the studied area the shallow seagrass beds are generally thriving in sedimentary zones). In Croatia, *Paracentrotus lividus* and/or *Arbacia lixula* overgrazing barren areas can be found frequently in very shallow waters (down to 5 m depth). This category was not included in the SL table, as it represents a low ecological status, but it seems to be potentially independent from the water quality and therefore sectors dominated by barren areas were excluded from the calculation of EQR values.

Using the information about the length of the coast occupied by a community type, the ecological quality value (EQ) was calculated using the formula:

$$EQ = \frac{\sum (l_i \times SL_i)}{\sum l_i}$$

EQ = ecological quality value of a coastline sector

l_i = length of the coastline with the community category i

SL_i = sensitivity level of the community category i

Table 1

The proposed list of macroalgal community types and their sensitivity levels (SL) in the eastern part of the Adriatic Sea.

Community type	Community description	Sensitivity level (SL)
<i>Cystoseira spicata</i> 3	Continuous belt of <i>Cystoseira amentacea</i> var. <i>spicata</i>	20
<i>Cystoseira crinitophylla</i>	Populations of <i>Cystoseira crinitophylla</i>	20
<i>Cystoseira crinita</i>	Populations of <i>Cystoseira crinita</i>	20
<i>Cystoseira corniculata</i>	Populations of <i>Cystoseira corniculata</i>	20
<i>Cystoseira foeniculacea</i>	Populations of <i>Cystoseira foeniculacea</i>	20
Trottoir	Organogenic build-ups of <i>Lithophyllum byssoides</i> and other coralline algae	20
<i>Cystoseira barbata</i>	Populations of <i>Cystoseira barbata</i> without other <i>Cystoseira</i> species	16
<i>Cystoseira spicata</i> 2	Abundant patches of <i>Cystoseira amentacea</i> var. <i>spicata</i>	15
<i>Cystoseira compressa</i>	Populations of <i>Cystoseira compressa</i> without other <i>Cystoseira</i> species	12
<i>Cystoseira spicata</i> 1	Rare scattered plants of <i>Cystoseira amentacea</i> var. <i>spicata</i>	10
Photophilic algae	Community dominated by <i>Padina</i> / <i>Dictyota</i> / <i>Dictyopteris</i> / <i>Taonia</i> / <i>Halopteris</i>	10
Corallina	Community dominated by <i>Corallina elongata</i> and/or <i>Haliptilon virgatum</i>	8
Mytilus	Community dominated by <i>Mytilus galloprovincialis</i>	6
Green algae	Community dominated by <i>Ulva</i> / <i>Enteromorpha</i> / <i>Cladophora</i>	3
Cyanobacteria	Cyanobacterial belt	1

Table 2

A list of geomorphological factors and their categories used to describe the coastline sectors in the Eastern Adriatic Sea.

Geomorphological factor	Category
Coast morphology	High coast Low coast Blocks
Substrate	Calcareous (limestone) Metamorphic Sandstone Conglomerate
Coastline slope	Horizontal (0–30°) Sub-vertical (30–60°) Vertical (60–90°) Overhanging
Coastline orientation	North Northeast East Southeast South Southwest West Northwest
Substrate rugosity	Smooth Rugose
Natural or artificial	Natural Artificial
Wave exposure	Sheltered Moderately exposed Very exposed

The establishment and abundance of natural shallow water macroalgal communities is known to be mainly determined by coastline geomorphology (Ercegović, 1964; Ballesteros, 1992) and such variability has to be taken into account to properly highlight the effects of potential human impacts. Following the original CARLIT method (Ballesteros et al., 2007), we quantified a set of geomorphological features potentially affecting the distribution of macroalgal communities along the Adriatic coastline (Table 2). The geomorphological features were discussed among the experts during the workshop cited above.

In order to calculate appropriate reference values for the Eastern Adriatic Sea, average EQ values were measured on sites with none or very minor anthropogenic impact. This is a common practice in the Mediterranean Geographical Inter-calibration Group (MedGIG) so we decided to follow the same impact levels used for other biological quality elements of the same working group:

- population density: less than 1000 inhabitants/km² in the next 15 km and/or less than 100 inhabitants/km² in the next 3 km within that area (winter population)
- less than 10% of artificial coastline
- no harbors (more than 100 boats) within 3 km
- no beach regeneration within 1 km
- no industries within 3 km
- no fish farms within 1 km

2.2. Data collection

We surveyed and applied CARLIT methodology to 100 km of coast in pristine areas of the Adriatic Sea, undisturbed by human activities, which included marine protected areas, remote islands and very sparsely inhabited coastal areas (Fig. 1A). The reference sites covered a wide range of geomorphological features and spanned from the north to the south, including coastal zone and islands. In order to validate the CARLIT application in the Eastern

Table 3

Types of pressure, Corine Land Cover (CLC) codes, area percentages, corresponding scores used in calculation of LUSI (Land Uses Simplified Index) and confinement correction factors used for pressure assessment in coastline sectors in the eastern part of the Adriatic Sea.

Type of pressure	CLC code	Area percentage (%)	Score
Urban area	11	<10 10–33 33–66 ≥66	0 1 2 3
Commercial and industrial area	12, 13	<10 10–30 ≥30	0 1 2
Agricultural area	21–24	<10 10–40 40–60 ≥60	0 1 2 3
Sewage outfall			1
Commercial harbor			1
Aquaculture			1
Freshwater input			1
Confinement correction factor	Concave Convex Straight		Multiply by 1.25 Multiply by 0.75 Multiply by 1.00

Adriatic, additional 200 km of rocky coast were surveyed, including urban, agricultural, industrial and harbor areas (Fig. 1A and B).

The entire assessed coastline was surveyed in late spring (June, 2010) with a rubber boat kept as close as possible to the shore. The homogenous sectors of the coast regarding upper-sublittoral communities (Table 1) and geomorphological factors (Table 2) were visually determined and noted directly into maps. The data were subsequently transferred into geo-referenced maps and the length of each sector was measured accurately. In order to determine the geomorphological factors which mostly influence the distribution of macroalgal communities, a BEST analysis (Clarke and Gorley, 2006) was performed on the percentages of coast (only considering sectors longer than 500 m) characterized by the combination of each geomorphological factor and community type inside reference areas. The number of output variables (geomorphological factors) in the BEST analysis was limited to two in order to obtain a reasonable number of geographical relevant situations (GRS) allowing an easy application of CARLIT method. A non-metric multidimensional scaling analysis (nMDS) was performed on the Bray–Curtis similarity matrix and the two most important geomorphological factors determined in the BEST analysis were represented with different symbols. For resulting GRS the average ecological quality value (EQ) was calculated.

2.3. Validation of results with human pressure data

The validation of results was performed on 27 sectors of coastline which were 5 km long and 6 sectors of coastline which were 500 m long, selected from the assessment of the whole coastline excluding the barren zones. In each of the selected sites, the calculated EQR values were compared with the values of LUSI (Land Uses Simplified Index), which measures human induced pressures on the coastal area according to the publicly available data (UNEP/MAP, 2011). Similar indices based on satellite-derived data on land use were proposed by Lopez y Royo et al. (2009) and Gardi et al. (2010). Surfaces of different land-use types (urban, industrial and agricultural) were extracted from the latest Corine Land Cover (2006) maps of Croatian coast spanning a 1 km wide strip along the 5 km long coastline sector where the method was applied. Scores were assigned to each type and level of pressure (Table 3) and added up to produce the final LUSI value. A correction factor was applied to

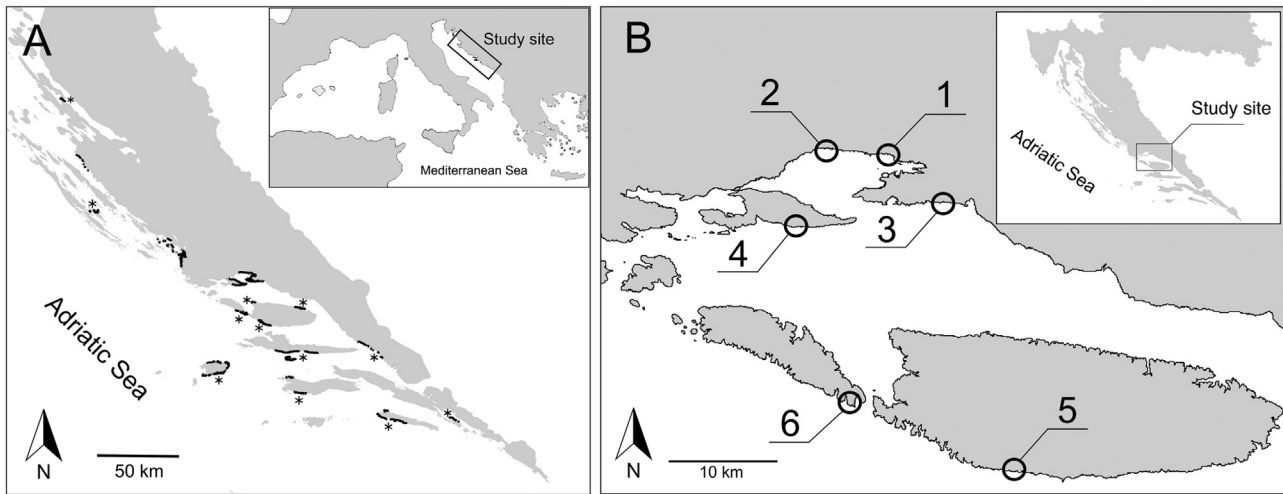


Fig. 1. Map of the investigated area showing: (A) sectors of coast where CARLIT was applied (bold line) and locations of reference areas (asterisk), (B) six stations in the supposed nutrient enrichment gradient where CARLIT was applied along with nutrient sampling.

the final value to account for the degree of confinement that could potentially emphasize the effect of pressures (UNEP/MAP, 2011).

In the six sites (500 m long), positioned along a supposed nutrient enrichment gradient, the EQR values were compared to nutrient concentrations (nitrates, nitrites, ammonia, orthophosphates and orthosilicates). The more polluted sites were positioned in the area of Kaštela Bay and Split city (sites 1 and 2, Fig. 1B) while the more pristine sites (5 and 6) included Šolta and Brač islands. The positions of sites 3 and 4 were intermediate to the positions of the aforementioned sites. The nutrient samples were taken in triplicate from the surface layer every 45 days during one year. Analyses were performed on SEAL AutoAnalyzer III using slightly modified automated methods (Grasshoff, 1976). A principal component analysis (PCA) was performed on the mean annual nutrient concentrations data and the EQR values of coastline sectors were compared to the values of the first axis of this PCA. PRIMER 6 software package (Clarke and Gorley, 2006) was used for statistical analyses.

3. Results

At the national scale, we surveyed 300 km of coastline (both reference and non reference sites). We observed that 39% of the surveyed coastline is populated by *Cystoseira* forests, mainly *Cystoseira amentacea* var. *spicata*. Photophilic algae without *Cystoseira* species were registered across 14% and coralline algae across 13% of the surveyed coastline. Green algae and mussel beds were found in less than 4% of the investigated area. Organogenic buildups of calcified algae (“trottoir”) occupied around 1% of the surveyed coastline, which underlines the importance of this rare community type in the Adriatic Sea. About 23% of the coastline was severely impacted by sea urchins *Paracentrotus lividus* and/or *Arbacia lixula* overgrazing. Locally, the percentage of affected coast ranged from 4% to over 85% in some geographical areas. In about 6% percent of the surveyed coastline CARLIT method was not applicable because it was sediment coastline or pebble beach.

The crucial step of CARLIT methodology implementation in the eastern coast of the Adriatic Sea was the calculation of reference values. Concerning the geomorphological analysis, the BEST analysis showed the highest correlation ($r=0.428$) of macroalgal community data with the geomorphological factors “coastal morphology” and “slope”. The combination of these two factors produced eight different “geomorphological relevant situations” (GRS). We added the category “blocks” to these eight GRS, because although it was not highly represented along Croatian coast, we

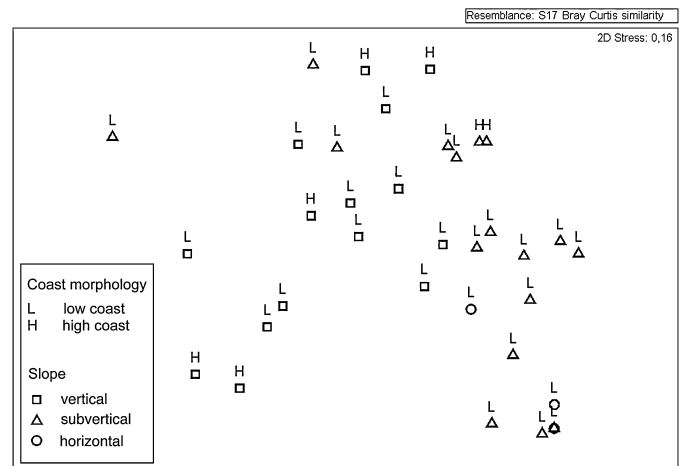


Fig. 2. nMDS analysis of geomorphological factors and associated macroalgal community data in pristine sectors of the Eastern Adriatic Sea. Symbols represent different levels of coast morphology (L – low coast; H – high coast) and slope (square – vertical; triangle – subvertical; circle – horizontal).

think this is relevant for distribution of communities. The results of the nMDS analysis performed on the reference sites data are reported in Fig. 2 and the EQ reference values calculated for each GRS are reported in Table 4.

The reference values were used to calculate EQR values in the 33 studied sites (27 sites of 5 km length and 6 sites of 500 m length). Results were significantly linearly related with LUSI index

Table 4

Average ecological quality values from the reference areas (EQ_{ref}) calculated for 9 geomorphological relevant situations (GRS) made by combination of the two most relevant factors (coast morphology and slope) that shape natural macroalgal communities in the Eastern Adriatic Sea.

GRS	Coast morphology	Coastline slope	EQ_{ref}
1	High coast	Horizontal	20.00
2	High coast	Sub-vertical	17.55
3	High coast	Vertical	12.96
4	High coast	Overhanging	10.00
5	Low coast	Horizontal	19.02
6	Low coast	Sub-vertical	17.72
7	Low coast	Vertical	14.62
8	Low coast	Overhanging	9.66
9	Blocks		12.76

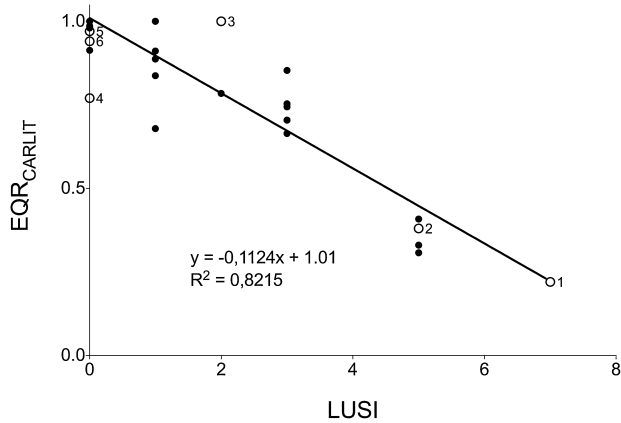


Fig. 3. Relation between the EQR values obtained by CARLIT method and corresponding LUSI values as a pressure assessment (5 km long random sectors – black circles, $n = 27$; 500 m long sectors along the supposed nutrient enrichment gradient – empty circles, $n = 6$).

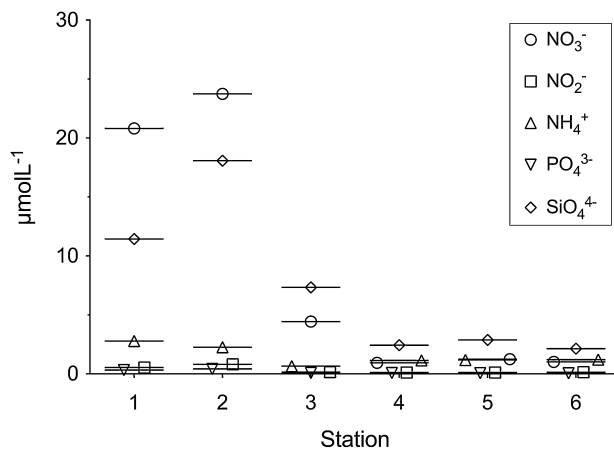


Fig. 4. Mean annual nutrient concentrations measured at six stations in the supposed nutrient enrichment gradient (NO_3^- – nitrates; NO_2^- – nitrites; NH_4^+ – ammonium salts; PO_4^{3-} – orthophosphates, SiO_4^{4-} – orthosilicates).

($R^2 = 0.8215$; $F = 142.6$; $DF = 31$, $P < 0.0001$) (Fig. 3) confirming a good response of the method to different levels of disturbance.

The studied nutrient enrichment gradient showed that the polluted sites (1 and 2) are effectively separated from the others (Figs. 4 and 5), which present nutrient values considered to be normal for the Adriatic coastal waters (IOF, 2011). The cumulative percentage of variation explained by the first two axes of the PCA is 99.2%, the first axis alone accounting for 90.3%. Ecological quality ratio (EQR) of each station calculated from CARLIT method application is presented in Fig. 6. The comparison of the

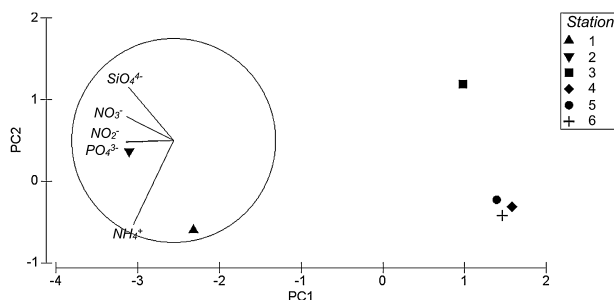


Fig. 5. PCA of mean annual nutrient concentrations measured in the supposed nutrient enrichment gradient.

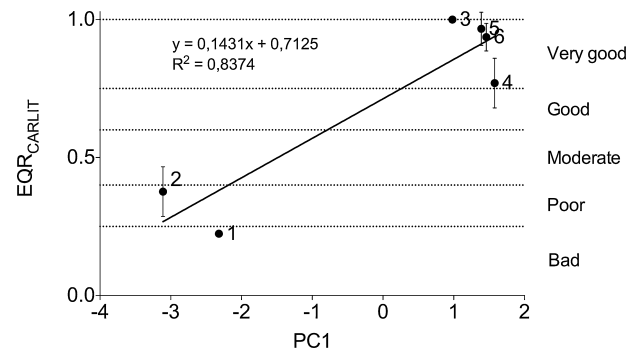


Fig. 6. Relation between the EQR values obtained by CARLIT method and values of first axis of PCA analysis of nutrient concentrations in the supposed nutrient enrichment gradient ($n = 6$) (ecological status classes are marked on the plot; error bars – standard error of the mean).

EQR values and values of first axis of PCA analysis of nutrient concentrations shows a good linear relation ($R^2 = 0.8374$; $F = 2060$; $DF = 4$, $P = 0.0105$) (Fig. 6).

4. Discussion

The coastline of Croatia (Eastern Adriatic Sea) measures 5835 km in length with more than 1000 islands and islets. In such a complex geographical environment, setting up a monitoring program to assess the ecological quality of coastal waters poses a major challenge. The sources of pollution are concentrated around big urban and industrial centers, but there are also many smaller sources of pollution, such as illegal sewage discharges, small harbors and marinas, aquaculture facilities and agricultural areas which are heterogeneously distributed (IOF, 2011). The heterogeneity in the sources of disturbance has been demonstrated as the most probable main source of uncertainty in assigning the ecological status to water bodies in Croatian coastal waters by the application of the *Posidonia oceanica* multivariate index (POMI) (Mascaró et al., 2012b). However, in the eastern coast of the Adriatic Sea, a multitude of pristine sites suitable for calculating reference values can still be found.

Our study revealed that *Cystoseira* forests thrive in 39% of the coastline. These brown algae are decreasing in several areas in the NW Mediterranean Sea (Thibaut et al., 2005) and are listed in the IUCN red list of macroalgae and seagrasses for Croatia (Antolić et al., 2011). Apart from the data presented here, no information is available about their conservation status in the Eastern Adriatic Sea. The collection of GIS data on the distribution and abundance of *Cystoseira* species in this country may represent a baseline for monitoring their evolution. This is particularly important when considering that about 23% of the coastline seems to be subjected to overgrazing by sea urchins, one of the major threats to Mediterranean marine algal forests. This process is generally associated with overfishing of sea urchin predators (Sala et al., 1998), but the reality is more complex (Cardona et al., 2007; Sala et al., 2012) and the very shallow barrens observed in Croatia in pristine areas should be linked to other processes. The inclusion of a “barren ground” category in the CARLIT methodology should be considered, but at present the sensitivity level that should be assigned to this category is still debated. Nevertheless, the monitoring of the barren areas coupled to the monitoring of *Cystoseira* forests would help in understanding the dynamics of shallow water communities and should represent a conservation priority.

The geomorphological factors mostly affecting macroalgal community distribution along Croatian coasts are coastal morphology and coastal slope. The surveyed coastline was predominantly a low coast type with horizontal to subvertical slopes. Ballesteros et al.

(2007) also found that the coastal morphology was a paramount factor, and as the second factor highlighted the nature of substrate (natural or artificial). This discrepancy may be due to the fact that artificial substrate is not commonly found in Croatia and probably is not relevant for the application of CARLIT in the studied area.

The EQR values recorded in this study show a good linear relation to LUSI index and nutrient concentrations as pressure indicators. This confirms that CARLIT method provides information about the pressures mainly connected with eutrophication and general degradation of water quality caused by coastal urbanization. Results of CARLIT application in Italy also showed a good relation to water quality variables (nutrients, water transparency and chlorophyll-*a*) and TRIX index (Mangialajo et al., 2007). The results of CARLIT application showed that a “high” and “good” ecological status can be assigned to the majority of coastal areas surveyed in this study, with the overall status of the whole assessed coastline classified as “high”. Only 9% of the whole coastline surveyed was classified in moderate, poor and bad ecological status. The “bad” status was recorded mostly in the northern part of Kaštela Bay, which is under strong anthropogenic influence, namely agricultural areas and untreated municipal or industrial wastewater discharges present in the Bay (IOF, 2011).

Martínez-Crego et al. (2010) have identified that most of the available biotic indices for assessing the status of coastal waters still fail in the broad-scale applicability, while CARLIT index, based on the cartography of the whole coastline (or an adequate hierarchical sampling) guarantees the representativeness of results on the broad scale. The CARLIT method has a low temporal variability, whilst a certain spatial variability was detected at the small scale, further underlining the importance of preferring an exhaustive cartography, whenever possible (Asnaghi et al., 2009). As it is highly unlikely to plan an exhaustive cartography of the Croatians coasts, the uncertainty of the method will be established on different spatial and temporal scales.

Considering that CARLIT raw data also allows to assess the evolution of communities with great conservation priority (e.g. *Cystoseira* forests) and that it is a non destructive method, we believe that CARLIT is a valuable tool for the implementation of the Water Framework Directive and other coastal zone management regulations like the Marine Strategy Framework Directive. The results give accurate spatial information on the quality and abundance of upper-sublittoral benthic communities and the associated ecological status, which can be efficiently used by managers and presented to the general public with the aid of advanced GIS tools.

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